

Cambridge IGCSE™

BIOLOGY**0610/61**

Paper 6 Alternative to Practical

October/November 2024

MARK SCHEME

Maximum Mark: 40

Published

This mark scheme is published as an aid to teachers and candidates, to indicate the requirements of the examination. It shows the basis on which Examiners were instructed to award marks. It does not indicate the details of the discussions that took place at an Examiners' meeting before marking began, which would have considered the acceptability of alternative answers.

Mark schemes should be read in conjunction with the question paper and the Principal Examiner Report for Teachers.

Cambridge International will not enter into discussions about these mark schemes.

Cambridge International is publishing the mark schemes for the October/November 2024 series for most Cambridge IGCSE, Cambridge International A and AS Level components, and some Cambridge O Level components.

This document consists of **8** printed pages.

PUBLISHED**Generic Marking Principles**

These general marking principles must be applied by all examiners when marking candidate answers. They should be applied alongside the specific content of the mark scheme or generic level descriptions for a question. Each question paper and mark scheme will also comply with these marking principles.

GENERIC MARKING PRINCIPLE 1:

Marks must be awarded in line with:

- the specific content of the mark scheme or the generic level descriptors for the question
- the specific skills defined in the mark scheme or in the generic level descriptors for the question
- the standard of response required by a candidate as exemplified by the standardisation scripts.

GENERIC MARKING PRINCIPLE 2:

Marks awarded are always **whole marks** (not half marks, or other fractions).

GENERIC MARKING PRINCIPLE 3:

Marks must be awarded **positively**:

- marks are awarded for correct/valid answers, as defined in the mark scheme. However, credit is given for valid answers which go beyond the scope of the syllabus and mark scheme, referring to your Team Leader as appropriate
- marks are awarded when candidates clearly demonstrate what they know and can do
- marks are not deducted for errors
- marks are not deducted for omissions
- answers should only be judged on the quality of spelling, punctuation and grammar when these features are specifically assessed by the question as indicated by the mark scheme. The meaning, however, should be unambiguous.

GENERIC MARKING PRINCIPLE 4:

Rules must be applied consistently, e.g. in situations where candidates have not followed instructions or in the application of generic level descriptors.

GENERIC MARKING PRINCIPLE 5:

Marks should be awarded using the full range of marks defined in the mark scheme for the question (however; the use of the full mark range may be limited according to the quality of the candidate responses seen).

GENERIC MARKING PRINCIPLE 6:

Marks awarded are based solely on the requirements as defined in the mark scheme. Marks should not be awarded with grade thresholds or grade descriptors in mind.

Science-Specific Marking Principles

- 1 Examiners should consider the context and scientific use of any keywords when awarding marks. Although keywords may be present, marks should not be awarded if the keywords are used incorrectly.
- 2 The examiner should not choose between contradictory statements given in the same question part, and credit should not be awarded for any correct statement that is contradicted within the same question part. Wrong science that is irrelevant to the question should be ignored.
- 3 Although spellings do not have to be correct, spellings of syllabus terms must allow for clear and unambiguous separation from other syllabus terms with which they may be confused (e.g. ethane / ethene, glucagon / glycogen, refraction / reflection).
- 4 The error carried forward (ecf) principle should be applied, where appropriate. If an incorrect answer is subsequently used in a scientifically correct way, the candidate should be awarded these subsequent marking points. Further guidance will be included in the mark scheme where necessary and any exceptions to this general principle will be noted.
- 5 'List rule' guidance

For questions that require ***n*** responses (e.g. State **two** reasons ...):
 - The response should be read as continuous prose, even when numbered answer spaces are provided.
 - Any response marked *ignore* in the mark scheme should not count towards ***n***.
 - Incorrect responses should not be awarded credit but will still count towards ***n***.
 - Read the entire response to check for any responses that contradict those that would otherwise be credited. Credit should **not** be awarded for any responses that are contradicted within the rest of the response. Where two responses contradict one another, this should be treated as a single incorrect response.
 - Non-contradictory responses after the first ***n*** responses may be ignored even if they include incorrect science.

6 Calculation specific guidance

Correct answers to calculations should be given full credit even if there is no working or incorrect working, **unless** the question states 'show your working'.

For questions in which the number of significant figures required is not stated, credit should be awarded for correct answers when rounded by the examiner to the number of significant figures given in the mark scheme. This may not apply to measured values.

For answers given in standard form (e.g. $a \times 10^n$) in which the convention of restricting the value of the coefficient (a) to a value between 1 and 10 is not followed, credit may still be awarded if the answer can be converted to the answer given in the mark scheme.

Unless a separate mark is given for a unit, a missing or incorrect unit will normally mean that the final calculation mark is not awarded. Exceptions to this general principle will be noted in the mark scheme.

7 Guidance for chemical equations

Multiples / fractions of coefficients used in chemical equations are acceptable unless stated otherwise in the mark scheme.

State symbols given in an equation should be ignored unless asked for in the question or stated otherwise in the mark scheme.

Mark scheme abbreviations

- ; separates marking points
- / alternative responses for the same marking point
- R reject the response
- A accept the response
- I ignore the response
- ecf error carried forward
- AVP any valid point
- ora or reverse argument
- AW alternative wording
- underline actual word given must be used by candidate (grammatical variants excepted)
- () the word / phrase in brackets is not required but sets the context

Question	Answer	Marks	Guidance
1(a)(i)	table drawn with minimum of three columns and a header line ; suitable headings for (test-)tube and distance with units in mm or cm ; recording of four correct measurements (for both S and W , each at 0 minutes and 15 minutes) ;	3	S ₀ , W ₀ , W ₁₅ = 53.5±2 mm, S ₁₅ = 65.0±2 mm
1(a)(ii)	correct calculation from candidate's results for S and W ;	1	
1(a)(iii)	water has moved into, bag S / the sucrose solution ;	1	
1(b)(i)	measuring cylinder / AVP ;	1	
1(b)(ii)	to remove any sucrose solution (from the outside of the dialysis tubing bag) ;	1	
1(b)(iii)	to ensure that it was the sucrose that was causing the effect / AW ;	1	
1(b)(iv)	<i>any two from:</i> volume of water in the test-tubes ; volume of liquid in the tubing bag / AW ; time (dialysis tubing bags left in, test-tubes / the water-bath) ; temperature (of the water-bath) ; type of dialysis, tubing / bag ; where the measurements were taken from ;	2	
1(b)(v)	to identify anomalous results / AW ;	1	
1(b)(vi)	<i>any one from:</i> measure the height of the liquid in the test-tube (at the start and end) ; measure the volume (inside the bag / in the test tube) ; measure the, mass of the liquid in bag / water in the test-tube) ;	1	
1(c)	add Benedict's (solution / reagent) ; heat ;	2	

Question	Answer	Marks	Guidance
2	<i>independent variable:</i> 1 at least two different temperatures ; <i>dependent variable:</i> 2 time for dye to reach the leaves or a set distance or distance moved by dye in a set time or number of cut sections containing dye ; 3 and 4 detail of method – max two from ;; • method of maintaining (at least one) air temperature • <i>idea of</i> slicing the end off the celery before immersion • cutting sections (with a, knife / scalpel) 5, 6 and 7 variables kept constant - max three from ;;; • wind-speed • humidity • light intensity • length / height / size, of, celery / stalk • number of leaves / surface area of leaves • species / type / age, of, celery • (named) colour / concentration / volume, of dye • time (in dye) / time stated • thickness of cut sections / sections cut in the same plane / AW 8 two or more replicates at each temperature / repeat the investigation at least two more times ; 9 suitable safety precaution ;	6	e.g. cutting sections on a flat or stable surface / cut away from body / gloves / goggles

Question	Answer	Marks	Guidance																											
3(a)	line AB = 90±1 (mm) ; 11 ;;	3	MP1 correct measurement of line AB MP2 correct calculation and answer to any number of significant figures MP3 correct rounding to two significant figures ecf from previous step																											
3(b)(i)	any three from: <table><tr><th>strawberry leaf</th><th></th><th>banana leaf</th></tr><tr><td>smaller</td><td>or</td><td>larger</td></tr><tr><td>has three leaflets /AW</td><td>or</td><td>one complete leaf</td></tr><tr><td>serrated / AW, leaf edge</td><td>or</td><td>smooth / AW, leaf edge</td></tr><tr><td>narrower / AW, vein</td><td>or</td><td>wider / AW, vein</td></tr><tr><td>leaf veins are at a steeper angle</td><td>or</td><td>leaf veins are at a less steep angle</td></tr><tr><td>branching vessels (between the side veins)</td><td>or</td><td>no branching vessels (between the side veins)</td></tr><tr><td>thinner, stalk / stem</td><td>or</td><td>thicker, stalk / stem</td></tr><tr><td>shorter, stalk / stem</td><td>or</td><td>longer, stalk / stem</td></tr></table> ;;;	strawberry leaf		banana leaf	smaller	or	larger	has three leaflets /AW	or	one complete leaf	serrated / AW, leaf edge	or	smooth / AW, leaf edge	narrower / AW, vein	or	wider / AW, vein	leaf veins are at a steeper angle	or	leaf veins are at a less steep angle	branching vessels (between the side veins)	or	no branching vessels (between the side veins)	thinner, stalk / stem	or	thicker, stalk / stem	shorter, stalk / stem	or	longer, stalk / stem	3	
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3(b)(ii)	limewater / AVP ;	1																												

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Question	Answer	Marks	Guidance										
3(c)	outline is a single clear unbroken line ; size is greater than half of space provided ; detail 1: rolled part (bract) shown ; detail 2: at least two overlapping bracts at the tip ;	4											
3(d)(i)	558 (g) ;;	2	MP1 correct calculation to any number of decimal places MP2 correct rounding to a whole number										
3(d)(ii)	axes labelled including units for protein content ; suitable linear scale on protein axis and fruit names on the other axis (in centre of bar) and bars occupy at least half the grid in both directions ; all bars plotted accurately $\pm\frac{1}{2}$ a small square ; bars of equal width and not touching and with equal space between the bars ;	4											
3(d)(iii)	<u>protein</u> content (of fruit) ;	1											
3(e)	<table><tr><td>type of food</td><td>protein test final colour</td><td>starch test final colour</td></tr><tr><td>X</td><td>purple</td><td>blue-black</td></tr><tr><td>Y</td><td>blue</td><td>blue-black</td></tr><tr><td>Z</td><td>purple</td><td>yellow-brown</td></tr></table> 	type of food	protein test final colour	starch test final colour	X	purple	blue-black	Y	blue	blue-black	Z	purple	yellow-brown
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